The fiber of the total map of a fiberwise transformation is the fiber of its component

The Univalent Foundations Program

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Editorial note. The counted statement is the labelled theorem ‘fibwise-fiber-total-fiber-equiv‘ (source lines 736-742, Theorem 4.7.6); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 4, section 7, theorem offset 5) make the counted theorem print as Theorem 4.7.6, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The internal label eq:uses-sum-over-paths is defined and used inside the retained proof, so it needs no external entry.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the fiber of a total map is the fiber of the component (Theorem 4.7.6).

Allowed standard prerequisites (editorial).

- Fiberwise transformation and its total map. A fiberwise transformation f : prd x:A, P(x) -> Q(x) induces a total map on the sigma types.. It is the map whose fiber is computed.
- Associativity of sigma types. Sigma types reassociate up to equivalence.. It is the first rewriting step at source line 750 and is used again at line 759.
- Paths in a sigma type. (a,u) = (x,v) unfolds into a path a = x together with a transported equality.. It is the step at source line 752.

Source reference. Original exercise 2.10: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1402–1402. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 2.7.2: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1853–1853. Original result cited by the retained

text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.11.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1844–1844. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.11.8: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1760–1760. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Theorem 4.7.6. *Suppose that f is a fiberwise transformation between families P and Q over a type A and let $x : A$ and $v : Q(x)$. Then we have an equivalence*

$$\mathsf{fib}_{\mathsf{total}(f)}((x, v)) \simeq \mathsf{fib}_{f(x)}(v).$$

Proof. We calculate:

$$\begin{aligned}
\mathsf{fib}_{\mathsf{total}(f)}((x,v)) &\equiv \sum_{w:\sum_{(x:A)} P(x)} (\mathsf{pr}_1 w, f(\mathsf{pr}_1 w, \mathsf{pr}_2 w)) = (x, v) \\
&\simeq \sum_{(a:A)} \sum_{(u:P(a))} (a, f(a,u)) = (x,v) \quad \text{(by original exercise 2.10)} \\
&\simeq \sum_{(a:A)} \sum_{(u:P(a))} \sum_{(p:a=x)} p_*(f(a,u)) = v \\
&\qquad\qquad \text{(by original theorem 2.7.2)} \\
&\simeq \sum_{(a:A)} \sum_{(p:a=x)} \sum_{(u:P(a))} p_*(f(a,u)) = v \\
&\simeq \sum_{u:P(x)} f(x,u) = v \qquad (*) \\
&\equiv \mathsf{fib}_{f(x)}(v).
\end{aligned}$$

The equivalence $(*)$ follows from original theorems 3.11.9 and 3.11.8 and original exercise 2.10. □